

FracFeedExtractor Sep 2025 – Jun 2026
ML pipeline that classifies ecological research
PDFs and extracts predator feeding-rate data.

DubBetter Ring Oct 2025
Real-time activity detection on a Raspberry Pi 5.

**Heart Disease
Predictor** Jun 2025 – Jul 2025
Heart disease risk model deployed to AWS and
Azure.

TalkTask Jan 2025 – Mar 2025
Voice-driven task manager with Gemini
recommendations.

BrainBurst Apr 2025
AI quiz generator built at a 24-hour hackathon.

Gym Database Simulation Apr 2025
PHP and SQL gym management app.

Small Shell Feb 2025
Linux shell in C with I/O redirection and signal
handling.

**Unlimited Games but No
Games** Dec 2024
Snake and Tetris prototypes on raylib.

FracFeedExtractor

Sep 2025 – Jun 2026

[Screenshot or architecture diagram]

End-to-end Python pipeline that classifies ecological research PDFs and extracts predator feeding-rate data into an analysis-ready database. Senior capstone with the OSU Novak Lab, team of four.

- [Your role and what you built]
- [Key technical decision]

94% classifier accuracy on a 234-paper held-out test set

Python

XGBoost

scikit-learn

Pydantic

PyMuPDF

Tesseract OCR

Ollama

pytest

GitHub Actions

View on GitHub ↗